

Newman-Penrose Formalism Calculations

Stephani et al., Equation (23.13a_2)

Metric

$$ds^2 = -\frac{1}{C(x)^2} dt \otimes dt + \frac{1}{C(x)^2} dx \otimes dx + C(x)^a S(t) dy \otimes dy + C(x)^a S(t) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$S = \sinh(2qt)$$

$$C = \cosh\left(\frac{2qx}{a}\right)$$

$$T = \tanh(qt)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}}{2C(x)}\right) dt + \left(\frac{\sqrt{2}}{2C(x)}\right) dx$$

$$n = \left(\frac{\sqrt{2}}{2C(x)}\right) dt + \left(-\frac{\sqrt{2}}{2C(x)}\right) dx$$

$$m = \left(\frac{1}{2} \sqrt{2} \sqrt{C(x)^a} \sqrt{S(t)} \right) dy + \left(\frac{1}{2} i \sqrt{2} \sqrt{C(x)^a} \sqrt{S(t)} \right) dz$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \sqrt{C(x)^a} \sqrt{S(t)} \right) dy + \left(-\frac{1}{2} i \sqrt{2} \sqrt{C(x)^a} \sqrt{S(t)} \right) dz$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} C(x) \right) dt + \left(\frac{1}{2} \sqrt{2} C(x) \right) dx$$

$$n = \left(-\frac{1}{2} \sqrt{2} C(x) \right) dt + \left(-\frac{1}{2} \sqrt{2} C(x) \right) dx$$

$$m = \left(\frac{\sqrt{2} \sqrt{|C(x)^a|} \sqrt{|S(t)|}}{2 C(x)^a S(t)} \right) dy + \left(\frac{i \sqrt{2} \sqrt{|C(x)^a|} \sqrt{|S(t)|}}{2 C(x)^a S(t)} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2} \sqrt{|C(x)^a|} \sqrt{|S(t)|}}{2 C(x)^a S(t)} \right) dy + \left(-\frac{i \sqrt{2} \sqrt{|C(x)^a|} \sqrt{|S(t)|}}{2 C(x)^a S(t)} \right) dz$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} C(x) \partial_t + \frac{1}{2} \sqrt{2} C(x) \partial_x$$

$$\Delta = -\frac{1}{2} \sqrt{2} C(x) \partial_t - \frac{1}{2} \sqrt{2} C(x) \partial_x$$

$$\delta = \frac{\sqrt{2} \sqrt{|C(x)^a|} \sqrt{|S(t)|}}{2 C(x)^a S(t)} \partial_y + \frac{i \sqrt{2} \sqrt{|C(x)^a|} \sqrt{|S(t)|}}{2 C(x)^a S(t)} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}\sqrt{|C(x)^a|\sqrt{|S(t)|}}}{2C(x)^a S(t)} \partial_y - \frac{i\sqrt{2}\sqrt{|C(x)^a|\sqrt{|S(t)|}}}{2C(x)^a S(t)} \partial_z$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)}{4C(x)^a|S(t)|} + \frac{\sqrt{2}C(x)|C(x)^a|\frac{\partial}{\partial t}S(t)}{4C(x)^a|S(t)|}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)}{4C(x)^a|S(t)|} - \frac{\sqrt{2}C(x)|C(x)^a|\frac{\partial}{\partial t}S(t)}{4C(x)^a|S(t)|}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{1}{8}\sqrt{2}a\frac{\partial}{\partial x}C(x) - \frac{\sqrt{2}a|C(x)^a||S(t)|\frac{\partial}{\partial x}C(x)}{8C(x)^aS(t)} - \frac{1}{4}\sqrt{2}\frac{\partial}{\partial x}C(x) - \frac{\sqrt{2}C(x)\frac{\partial}{\partial t}S(t)}{8S(t)} + \frac{\sqrt{2}C(x)|C(x)^a||S(t)|\frac{\partial}{\partial t}S(t)}{8C(x)^aS(t)^2}$$

$$\gamma = \frac{1}{8} \sqrt{2} a \frac{\partial}{\partial x} C(x) - \frac{\sqrt{2} a |C(x)^a| |S(t)| \frac{\partial}{\partial x} C(x)}{8 C(x)^a S(t)} - \frac{1}{4} \sqrt{2} \frac{\partial}{\partial x} C(x) + \frac{\sqrt{2} C(x) \frac{\partial}{\partial t} S(t)}{8 S(t)} - \frac{\sqrt{2} C(x) |C(x)^a| |S(t)| \frac{\partial}{\partial t} S(t)}{8 C(x)^a S(t)^2}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{1}{8} a^2 \frac{\partial}{\partial x} C(x)^2 - \frac{1}{4} a \frac{\partial}{\partial x} C(x)^2 - \frac{1}{4} a C(x) \frac{\partial^2}{(\partial x)^2} C(x) + \frac{a C(x) \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{4 S(t)} + \frac{C(x) \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{2 S(t)} + \frac{C(x)^2 \frac{\partial}{\partial t} S(t)^2}{8 S(t)^2} - \frac{C(x)^2 \frac{\partial^2}{(\partial t)^2} S(t)}{4 S(t)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned} \Phi_{11} = & -\frac{1}{32} a^2 \frac{\partial}{\partial x} C(x)^2 - \frac{a^2 S(t) |C(x)^a| \frac{\partial}{\partial x} C(x)^2}{32 C(x)^a |S(t)|} + \frac{S(t) |C(x)^a| \frac{\partial}{\partial x} C(x)^2}{8 C(x)^a |S(t)|} + \frac{1}{8} \frac{\partial}{\partial x} C(x)^2 \\ & - \frac{1}{8} C(x) \frac{\partial^2}{(\partial x)^2} C(x) - \frac{C(x) S(t) |C(x)^a| \frac{\partial^2}{(\partial x)^2} C(x)}{8 C(x)^a |S(t)|} + \frac{C(x)^2 \frac{\partial}{\partial t} S(t)^2}{32 S(t)^2} + \frac{C(x)^2 |C(x)^a| \frac{\partial}{\partial t} S(t)^2}{32 C(x)^a S(t) |S(t)|} \end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{1}{8} a^2 \frac{\partial}{\partial x} C(x)^2 - \frac{1}{4} a \frac{\partial}{\partial x} C(x)^2 - \frac{1}{4} a C(x) \frac{\partial^2}{(\partial x)^2} C(x) - \frac{a C(x) \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{4 S(t)} - \frac{C(x) \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{2 S(t)} + \frac{C(x)^2 \frac{\partial}{\partial t} S(t)^2}{8 S(t)^2} - \frac{C(x)^2 \frac{\partial^2}{(\partial t)^2} S(t)}{4 S(t)}$$

$$\Lambda = -\frac{1}{16}a^2\frac{\partial}{\partial x}C(x)^2 + \frac{1}{12}a\frac{\partial}{\partial x}C(x)^2 - \frac{1}{12}aC(x)\frac{\partial^2}{(\partial x)^2}C(x) - \frac{1}{12}\frac{\partial}{\partial x}C(x)^2 + \frac{1}{12}C(x)\frac{\partial^2}{(\partial x)^2}C(x) - \frac{C(x)^2\frac{\partial}{\partial t}S(t)^2}{48S(t)^2} + \frac{C(x)^2\frac{\partial^2}{(\partial t)^2}S(t)}{12S(t)}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{12C(x)^a|S(t)|} - \frac{aC(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{12C(x)^a|S(t)|} + \frac{S(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{6C(x)^a|S(t)|} - \frac{C(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{6C(x)^a|S(t)|} - \frac{C(x)^2|C(x)^a|\frac{\partial}{\partial t}S(t)^2}{12C(x)^aS(t)|S(t)|} + \frac{C(x)^2|C(x)^a|\frac{\partial^2}{(\partial t)^2}S(t)}{12C(x)^a|S(t)|}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

$$\begin{aligned} NP1 = & \frac{a^2|C(x)^a|^2\frac{\partial}{\partial x}C(x)^2}{4C(x)^{2a}} - \frac{3a^2C(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{8C(x)^{2a}|S(t)|} + \frac{a^2C(x)S(t)\overline{C(x)^a\frac{\partial}{\partial x}C(x)}\frac{\partial}{\partial x}C(x)}{8|C(x)^a||S(t)|\overline{C(x)}} + \frac{aC(x)^aC(x)|C(x)^a|\overline{\frac{\partial}{\partial t}S(t)}\frac{\partial}{\partial x}C(x)}{8C(x)^{2a}|S(t)|} \\ & - \frac{1}{8}a^2\frac{\partial}{\partial x}C(x)^2 + \frac{aC(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{4C(x)^{2a}|S(t)|} + \frac{a^2S(t)\overline{C(x)^a\frac{\partial}{\partial x}C(x)}\frac{\partial}{\partial x}C(x)^2}{8|C(x)^a||S(t)|} + \frac{aC(x)^aC(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{4C(x)^{2a}|S(t)|} - \frac{aC(x)|C(x)^a|^2\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{2C(x)^{2a}S(t)} \\ & + \frac{aC(x)^aC(x)|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{4C(x)^{2a}|S(t)|} + \frac{aC(x)^aC(x)|C(x)^a|\overline{S(t)}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{8C(x)^{2a}S(t)|S(t)|} - \frac{1}{4}a\frac{\partial}{\partial x}C(x)^2 - \frac{1}{4}aC(x)\frac{\partial^2}{(\partial x)^2}C(x) - \frac{aC(x)^2\overline{C(x)^a\frac{\partial}{\partial x}C(x)}\frac{\partial}{\partial t}S(t)}{8|C(x)^a||S(t)|\overline{C(x)}} \\ & - \frac{C(x)^aC(x)^2|C(x)^a|\frac{\partial}{\partial t}S(t)\frac{\partial}{\partial t}S(t)}{8C(x)^{2a}S(t)|S(t)|} + \frac{aC(x)\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{4S(t)} - \frac{C(x)^aC(x)|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{2C(x)^{2a}|S(t)|} - \frac{aC(x)\overline{C(x)^a\frac{\partial}{\partial x}C(x)}\frac{\partial}{\partial t}S(t)}{8|C(x)^a||S(t)|} + \frac{C(x)^2|C(x)^a|^2\frac{\partial}{\partial t}S(t)^2}{4C(x)^{2a}S(t)^2} \\ & - \frac{C(x)^aC(x)^2|C(x)^a|\frac{\partial}{\partial t}S(t)^2}{8C(x)^{2a}S(t)|S(t)|} - \frac{C(x)^aC(x)^2|C(x)^a|\overline{S(t)}\frac{\partial}{\partial t}S(t)^2}{8C(x)^{2a}S(t)^2|S(t)|} + \frac{C(x)^aC(x)^2|C(x)^a|\frac{\partial^2}{(\partial t)^2}S(t)}{4C(x)^{2a}|S(t)|} + \frac{C(x)\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{2S(t)} + \frac{C(x)^2\frac{\partial}{\partial t}S(t)^2}{8S(t)^2} - \frac{C(x)^2\frac{\partial^2}{(\partial t)^2}S(t)}{4S(t)} \end{aligned}$$

$$\begin{aligned}
NP6 = & -\frac{a^2|C(x)^a|^2\frac{\partial}{\partial x}C(x)^2}{8C(x)^{2a}} - \frac{a^2C(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{32C(x)^{2a}|S(t)|} + \frac{a^2C(x)S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial x}C(x)}{8|C(x)^a||S(t)|\overline{C(x)}} - \frac{3}{32}a^2\frac{\partial}{\partial x}C(x)^2 - \frac{5aC(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{12C(x)^{2a}|S(t)|} \\
& + \frac{a^2S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)^2}{8|C(x)^a||S(t)|} + \frac{aC(x)^aC(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{6C(x)^{2a}|S(t)|} + \frac{5}{12}a\frac{\partial}{\partial x}C(x)^2 + \frac{7C(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{24C(x)^{2a}|S(t)|} - \frac{1}{6}aC(x)\frac{\partial^2}{(\partial x)^2}C(x) \\
& - \frac{7C(x)^aC(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{24C(x)^{2a}|S(t)|} - \frac{C(x)^aC(x)^2|C(x)^a|\frac{\partial}{\partial t}S(t)\frac{\partial}{\partial t}S(t)}{8C(x)^{2a}S(t)|S(t)|} + \frac{C(x)^2|C(x)^a|^2\frac{\partial}{\partial t}S(t)^2}{8C(x)^{2a}S(t)^2} + \frac{19C(x)^aC(x)^2|C(x)^a|\frac{\partial}{\partial t}S(t)^2}{96C(x)^{2a}S(t)|S(t)|} \\
& - \frac{C(x)^aC(x)^2|C(x)^a|\overline{S(t)}\frac{\partial}{\partial t}S(t)^2}{8C(x)^{2a}S(t)^2|S(t)|} - \frac{C(x)^aC(x)^2|C(x)^a|\frac{\partial^2}{(\partial t)^2}S(t)}{6C(x)^{2a}|S(t)|} - \frac{7}{24}\frac{\partial}{\partial x}C(x)^2 + \frac{7}{24}C(x)\frac{\partial^2}{(\partial x)^2}C(x) - \frac{7C(x)^2\frac{\partial}{\partial t}S(t)^2}{96S(t)^2} + \frac{C(x)^2\frac{\partial^2}{(\partial t)^2}S(t)}{6S(t)}
\end{aligned}$$

$$\begin{aligned}
NP8 = & \frac{a^2C(x)S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial x}C(x)}{8|C(x)^a||S(t)|\overline{C(x)}} - \frac{1}{8}a^2\frac{\partial}{\partial x}C(x)^2 - \frac{a^2S(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{8C(x)^a|S(t)|} + \frac{a^2S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)^2}{8|C(x)^a||S(t)|} + \frac{aC(x)|C(x)^a|\frac{\partial}{\partial t}S(t)\frac{\partial}{\partial x}C(x)}{8C(x)^a|S(t)|} + \frac{1}{6}a\frac{\partial}{\partial x}C(x)^2 \\
& - \frac{aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{6C(x)^a|S(t)|} - \frac{1}{6}aC(x)\frac{\partial^2}{(\partial x)^2}C(x) + \frac{aC(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{6C(x)^a|S(t)|} + \frac{aC(x)^2\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{8|C(x)^a||S(t)|\overline{C(x)}} - \frac{aC(x)|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{2C(x)^a|S(t)|} \\
& + \frac{aC(x)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{8|C(x)^a||S(t)|} + \frac{aC(x)|C(x)^a|\overline{S(t)}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)}{8C(x)^aS(t)|S(t)|} + \frac{S(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{6C(x)^a|S(t)|} - \frac{1}{6}\frac{\partial}{\partial x}C(x)^2 + \frac{1}{6}C(x)\frac{\partial^2}{(\partial x)^2}C(x) - \frac{C(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{6C(x)^a|S(t)|} \\
& + \frac{C(x)^2|C(x)^a|\frac{\partial}{\partial t}S(t)\frac{\partial}{\partial t}S(t)}{8C(x)^aS(t)|S(t)|} - \frac{C(x)^2\frac{\partial}{\partial t}S(t)^2}{24S(t)^2} - \frac{5C(x)^2|C(x)^a|\frac{\partial}{\partial t}S(t)^2}{24C(x)^aS(t)|S(t)|} + \frac{C(x)^2|C(x)^a|\overline{S(t)}\frac{\partial}{\partial t}S(t)^2}{8C(x)^aS(t)^2|S(t)|} + \frac{C(x)^2\frac{\partial^2}{(\partial t)^2}S(t)}{6S(t)} - \frac{C(x)^2|C(x)^a|\frac{\partial^2}{(\partial t)^2}S(t)}{6C(x)^a|S(t)|}
\end{aligned}$$

$$\begin{aligned}
NP12 = & \frac{a^2|C(x)^a|^2\frac{\partial}{\partial x}C(x)^2}{8C(x)^{2a}} - \frac{a^2C(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{32C(x)^{2a}|S(t)|} - \frac{3}{32}a^2\frac{\partial}{\partial x}C(x)^2 - \frac{aC(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{12C(x)^{2a}|S(t)|} + \frac{aC(x)^aC(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{12C(x)^{2a}|S(t)|} \\
& + \frac{1}{12}a\frac{\partial}{\partial x}C(x)^2 - \frac{C(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^2}{24C(x)^{2a}|S(t)|} - \frac{1}{12}aC(x)\frac{\partial^2}{(\partial x)^2}C(x) + \frac{C(x)^aC(x)S(t)|C(x)^a|\frac{\partial^2}{(\partial x)^2}C(x)}{24C(x)^{2a}|S(t)|} - \frac{C(x)^2|C(x)^a|^2\frac{\partial}{\partial t}S(t)^2}{8C(x)^{2a}S(t)^2} \\
& + \frac{11C(x)^aC(x)^2|C(x)^a|\frac{\partial}{\partial t}S(t)^2}{96C(x)^{2a}S(t)|S(t)|} - \frac{C(x)^aC(x)^2|C(x)^a|\frac{\partial^2}{(\partial t)^2}S(t)}{12C(x)^{2a}|S(t)|} + \frac{1}{24}\frac{\partial}{\partial x}C(x)^2 - \frac{1}{24}C(x)\frac{\partial^2}{(\partial x)^2}C(x) + \frac{C(x)^2\frac{\partial}{\partial t}S(t)^2}{96S(t)^2} + \frac{C(x)^2\frac{\partial^2}{(\partial t)^2}S(t)}{12S(t)}
\end{aligned}$$

$$\begin{aligned}
NP14 = & \frac{a^2 |C(x)^a|^2 \frac{\partial}{\partial x} C(x)^2}{4 C(x)^{2a}} - \frac{3 a^2 C(x)^a S(t) |C(x)^a| \frac{\partial}{\partial x} C(x)^2}{8 C(x)^{2a} |S(t)|} + \frac{a^2 C(x) S(t) \overline{C(x)^a \frac{\partial}{\partial x} C(x)} \frac{\partial}{\partial x} C(x)}{8 |C(x)^a| |S(t) \overline{C(x)}|} - \frac{a C(x)^a C(x) |C(x)^a| \overline{\frac{\partial}{\partial t} S(t)} \frac{\partial}{\partial x} C(x)}{8 C(x)^{2a} |S(t)|} \\
& - \frac{1}{8} a^2 \frac{\partial}{\partial x} C(x)^2 + \frac{a C(x)^a S(t) |C(x)^a| \frac{\partial}{\partial x} C(x)^2}{4 C(x)^{2a} |S(t)|} + \frac{a^2 S(t) \overline{C(x)^a \frac{\partial}{\partial x} C(x)}^2}{8 |C(x)^a| |S(t)|} + \frac{a C(x)^a C(x) S(t) |C(x)^a| \frac{\partial^2}{(\partial x)^2} C(x)}{4 C(x)^{2a} |S(t)|} + \frac{a C(x) |C(x)^a|^2 \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{2 C(x)^{2a} S(t)} \\
& - \frac{a C(x)^a C(x) |C(x)^a| \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{4 C(x)^{2a} |S(t)|} - \frac{a C(x)^a C(x) |C(x)^a| \overline{S(t)} \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{8 C(x)^{2a} S(t) |S(t)|} - \frac{1}{4} a \frac{\partial}{\partial x} C(x)^2 - \frac{1}{4} a C(x) \frac{\partial^2}{(\partial x)^2} C(x) + \frac{a C(x)^2 \overline{C(x)^a \frac{\partial}{\partial x} C(x)} \frac{\partial}{\partial t} S(t)}{8 |C(x)^a| |S(t) \overline{C(x)}|} \\
& - \frac{C(x)^a C(x)^2 |C(x)^a| \frac{\partial}{\partial t} S(t) \frac{\partial}{\partial t} S(t)}{8 C(x)^{2a} S(t) |S(t)|} - \frac{a C(x) \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{4 S(t)} + \frac{C(x)^a C(x) |C(x)^a| \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{2 C(x)^{2a} |S(t)|} + \frac{a C(x) \overline{C(x)^a \frac{\partial}{\partial x} C(x)} \frac{\partial}{\partial t} S(t)}{8 |C(x)^a| |S(t)|} + \frac{C(x)^2 |C(x)^a|^2 \frac{\partial}{\partial t} S(t)^2}{4 C(x)^{2a} S(t)^2} \\
& - \frac{C(x)^a C(x)^2 |C(x)^a| \frac{\partial}{\partial t} S(t)^2}{8 C(x)^{2a} S(t) |S(t)|} - \frac{C(x)^a C(x)^2 |C(x)^a| \overline{S(t)} \frac{\partial}{\partial t} S(t)^2}{8 C(x)^{2a} S(t)^2 |S(t)|} + \frac{C(x)^a C(x)^2 |C(x)^a| \frac{\partial^2}{(\partial t)^2} S(t)}{4 C(x)^{2a} |S(t)|} - \frac{C(x) \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{2 S(t)} + \frac{C(x)^2 \frac{\partial}{\partial t} S(t)^2}{8 S(t)^2} - \frac{C(x)^2 \frac{\partial^2}{(\partial t)^2} S(t)}{4 S(t)}
\end{aligned}$$

$$\begin{aligned}
NP17 = & - \frac{a^2 C(x) S(t) \overline{C(x)^a \frac{\partial}{\partial x} C(x)} \frac{\partial}{\partial x} C(x)}{8 |C(x)^a| |S(t) \overline{C(x)}|} + \frac{1}{8} a^2 \frac{\partial}{\partial x} C(x)^2 + \frac{a^2 S(t) |C(x)^a| \frac{\partial}{\partial x} C(x)^2}{8 C(x)^a |S(t)|} - \frac{a^2 S(t) \overline{C(x)^a \frac{\partial}{\partial x} C(x)}^2}{8 |C(x)^a| |S(t)|} + \frac{a C(x) |C(x)^a| \overline{\frac{\partial}{\partial t} S(t)} \frac{\partial}{\partial x} C(x)}{8 C(x)^a |S(t)|} - \frac{1}{6} a \frac{\partial}{\partial x} C(x)^2 \\
& + \frac{a S(t) |C(x)^a| \frac{\partial}{\partial x} C(x)^2}{6 C(x)^a |S(t)|} + \frac{1}{6} a C(x) \frac{\partial^2}{(\partial x)^2} C(x) - \frac{a C(x) S(t) |C(x)^a| \frac{\partial^2}{(\partial x)^2} C(x)}{6 C(x)^a |S(t)|} + \frac{a C(x)^2 \overline{C(x)^a \frac{\partial}{\partial x} C(x)} \frac{\partial}{\partial t} S(t)}{8 |C(x)^a| |S(t) \overline{C(x)}|} - \frac{a C(x) |C(x)^a| \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{2 C(x)^a |S(t)|} \\
& + \frac{a C(x) \overline{C(x)^a \frac{\partial}{\partial x} C(x)} \frac{\partial}{\partial t} S(t)}{8 |C(x)^a| |S(t)|} + \frac{a C(x) |C(x)^a| \overline{S(t)} \frac{\partial}{\partial x} C(x) \frac{\partial}{\partial t} S(t)}{8 C(x)^a S(t) |S(t)|} - \frac{S(t) |C(x)^a| \frac{\partial}{\partial x} C(x)^2}{6 C(x)^a |S(t)|} + \frac{1}{6} \frac{\partial}{\partial x} C(x)^2 - \frac{1}{6} C(x) \frac{\partial^2}{(\partial x)^2} C(x) + \frac{C(x) S(t) |C(x)^a| \frac{\partial^2}{(\partial x)^2} C(x)}{6 C(x)^a |S(t)|} \\
& - \frac{C(x)^2 |C(x)^a| \frac{\partial}{\partial t} S(t) \frac{\partial}{\partial t} S(t)}{8 C(x)^a S(t) |S(t)|} + \frac{C(x)^2 \frac{\partial}{\partial t} S(t)^2}{24 S(t)^2} + \frac{5 C(x)^2 |C(x)^a| \frac{\partial}{\partial t} S(t)^2}{24 C(x)^a S(t) |S(t)|} - \frac{C(x)^2 |C(x)^a| \overline{S(t)} \frac{\partial}{\partial t} S(t)^2}{8 C(x)^a S(t)^2 |S(t)|} - \frac{C(x)^2 \frac{\partial^2}{(\partial t)^2} S(t)}{6 S(t)} + \frac{C(x)^2 |C(x)^a| \frac{\partial^2}{(\partial t)^2} S(t)}{6 C(x)^a |S(t)|}
\end{aligned}$$

Equations are vanished.

Bianchi Identities

$$\begin{aligned}
BI9 = & \frac{\sqrt{2}a^3|C(x)^a|^2\frac{\partial}{\partial x}C(x)^3}{32C(x)^{2a}} + \frac{\sqrt{2}a^3C(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^3}{64C(x)^{2a}|S(t)|} + \frac{\sqrt{2}a^3C(x)S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial x}C(x)^2}{128|C(x)^a||S(t)|\overline{C(x)}} + \frac{\sqrt{2}a^2C(x)^aC(x)|C(x)^a|\frac{\partial}{\partial t}S(t)\frac{\partial}{\partial x}C(x)^2}{128C(x)^{2a}|S(t)|} \\
& - \frac{1}{16}\sqrt{2}a^3\frac{\partial}{\partial x}C(x)^3 + \frac{\sqrt{2}a^3S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)^3}{128|C(x)^a||S(t)|} + \frac{\sqrt{2}a^2C(x)^aC(x)S(t)|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x)}{32C(x)^{2a}|S(t)|} - \frac{\sqrt{2}a^2C(x)|C(x)^a|^2\frac{\partial}{\partial x}C(x)^2\frac{\partial}{\partial t}S(t)}{32C(x)^{2a}S(t)} \\
& - \frac{3\sqrt{2}a^2C(x)^aC(x)|C(x)^a|\frac{\partial}{\partial x}C(x)^2\frac{\partial}{\partial t}S(t)}{64C(x)^{2a}|S(t)|} + \frac{\sqrt{2}a^2C(x)^aC(x)|C(x)^a|\overline{S(t)}\frac{\partial}{\partial x}C(x)^2\frac{\partial}{\partial t}S(t)}{128C(x)^{2a}S(t)|S(t)|} - \frac{\sqrt{2}a|C(x)^a|^2\frac{\partial}{\partial x}C(x)^3}{8C(x)^{2a}} - \frac{\sqrt{2}aC(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^3}{16C(x)^{2a}|S(t)|} \\
& - \frac{1}{32}\sqrt{2}a^2C(x)\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x) + \frac{\sqrt{2}aC(x)|C(x)^a|^2\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x)}{8C(x)^{2a}} + \frac{\sqrt{2}aC(x)^aC(x)S(t)|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x)}{16C(x)^{2a}|S(t)|} \\
& + \frac{\sqrt{2}a^2C(x)\frac{\partial}{\partial x}C(x)^2\frac{\partial}{\partial t}S(t)}{16S(t)} - \frac{\sqrt{2}aC(x)^2|C(x)^a|^2\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)^2}{32C(x)^{2a}S(t)^2} - \frac{\sqrt{2}aC(x)^aC(x)^2|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)^2}{64C(x)^{2a}S(t)|S(t)|} - \frac{\sqrt{2}aC(x)S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial x}C(x)^2}{32|C(x)^a||S(t)|\overline{C(x)}} \\
& - \frac{\sqrt{2}C(x)^aC(x)|C(x)^a|\frac{\partial}{\partial t}S(t)\frac{\partial}{\partial x}C(x)^2}{32C(x)^{2a}|S(t)|} + \frac{1}{4}\sqrt{2}a\frac{\partial}{\partial x}C(x)^3 - \frac{\sqrt{2}aS(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)^3}{32|C(x)^a||S(t)|} + \frac{\sqrt{2}aC(x)^2S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x)}{32|C(x)^a||S(t)|\overline{C(x)}} + \dots
\end{aligned}$$

$$\begin{aligned}
BI11 = & -\frac{\sqrt{2}a^3|C(x)^a|^2\frac{\partial}{\partial x}C(x)^3}{32C(x)^{2a}} - \frac{\sqrt{2}a^3C(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^3}{64C(x)^{2a}|S(t)|} - \frac{\sqrt{2}a^3C(x)S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial x}C(x)^2}{128|C(x)^a||S(t)|\overline{C(x)}} + \frac{\sqrt{2}a^2C(x)^aC(x)|C(x)^a|\frac{\partial}{\partial t}S(t)\frac{\partial}{\partial x}C(x)^2}{128C(x)^{2a}|S(t)|} \\
& + \frac{1}{16}\sqrt{2}a^3\frac{\partial}{\partial x}C(x)^3 - \frac{\sqrt{2}a^3S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)^3}{128|C(x)^a||S(t)|} - \frac{\sqrt{2}a^2C(x)^aC(x)S(t)|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x)}{32C(x)^{2a}|S(t)|} - \frac{\sqrt{2}a^2C(x)|C(x)^a|^2\frac{\partial}{\partial x}C(x)^2\frac{\partial}{\partial t}S(t)}{32C(x)^{2a}S(t)} \\
& - \frac{3\sqrt{2}a^2C(x)^aC(x)|C(x)^a|\frac{\partial}{\partial x}C(x)^2\frac{\partial}{\partial t}S(t)}{64C(x)^{2a}|S(t)|} + \frac{\sqrt{2}a^2C(x)^aC(x)|C(x)^a|\overline{S(t)}\frac{\partial}{\partial x}C(x)^2\frac{\partial}{\partial t}S(t)}{128C(x)^{2a}S(t)|S(t)|} + \frac{\sqrt{2}a|C(x)^a|^2\frac{\partial}{\partial x}C(x)^3}{8C(x)^{2a}} + \frac{\sqrt{2}aC(x)^aS(t)|C(x)^a|\frac{\partial}{\partial x}C(x)^3}{16C(x)^{2a}|S(t)|} \\
& + \frac{1}{32}\sqrt{2}a^2C(x)\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x) - \frac{\sqrt{2}aC(x)|C(x)^a|^2\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x)}{8C(x)^{2a}} - \frac{\sqrt{2}aC(x)^aC(x)S(t)|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x)}{16C(x)^{2a}|S(t)|} \\
& + \frac{\sqrt{2}a^2C(x)\frac{\partial}{\partial x}C(x)^2\frac{\partial}{\partial t}S(t)}{16S(t)} + \frac{\sqrt{2}aC(x)^2|C(x)^a|^2\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)^2}{32C(x)^{2a}S(t)^2} + \frac{\sqrt{2}aC(x)^aC(x)^2|C(x)^a|\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial t}S(t)^2}{64C(x)^{2a}S(t)|S(t)|} + \frac{\sqrt{2}aC(x)S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial}{\partial x}C(x)^2}{32|C(x)^a||S(t)|\overline{C(x)}} \\
& - \frac{\sqrt{2}C(x)^aC(x)|C(x)^a|\frac{\partial}{\partial t}S(t)\frac{\partial}{\partial x}C(x)^2}{32C(x)^{2a}|S(t)|} - \frac{1}{4}\sqrt{2}a\frac{\partial}{\partial x}C(x)^3 + \frac{\sqrt{2}aS(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)^3}{32|C(x)^a||S(t)|} - \frac{\sqrt{2}aC(x)^2S(t)\overline{C(x)^a}\frac{\partial}{\partial x}C(x)\frac{\partial^2}{(\partial x)^2}C(x)}{32|C(x)^a||S(t)|\overline{C(x)}} + \dots
\end{aligned}$$

Equations are vanished.

Petrov Type

Type "D"